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Cyber Attack Detection Scheme in Smart Grid

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1. Introduction

Smart grids use historical and real-time data for dynamic optimization of grid operation and resource utilization. With the increasing use of advanced communication systems and measurement devices, smart grids are vulnerable to cyber threats.

One of the most dangerous forms of cyber-attacks is the **False Data Injection (FDI)** attack, where an attacker manipulates measurement data to mislead state estimation without being detected.

This project presents an approach to detect FDI-based cyber attacks using the **Kalman Filter** technique, comparing it with the conventional Weighted Least Squares (WLS) method.

Key Features of Smart Grid Cybersecurity

- Integration of sensors, smart meters, and communication channels.
- Exposure to cyber-physical threats.
- Requirement for real-time monitoring and estimation accuracy.

2. Objective

The objectives of this work are:

1. To verify how a stealth FDI attack can bypass the residual-based detection in the WLS method.
2. To design a Kalman Filter-based approach to detect stealth FDI-based cyber attacks.
3. To analyze bus voltages before and after the cyber attack.

3. Why Kalman Filter Instead of Traditional Residual Method?

The Kalman Filter is a recursive algorithm that estimates the system state by combining predictions from the system model with new measurements.

Advantages of Kalman Filter

- Effectively handles noise and uncertainty.
- Captures dynamic behavior of the power system.

- Recursive nature enables real-time operation.
- Statistically reliable — handles covariance error and provides confidence bounds for detection.

4. Kalman Filter Concept

The Kalman Filter integrates three key elements:

- **System Model:** Describes state evolution.
- **Measurement Model:** Relates system states to observable outputs.
- **Noise Models:** Represent process noise (w_k) and measurement noise (v_k).

Mathematical Formulation

$$x_{k+1} = Ax_k + Bu_k + w_k \quad (1)$$

$$z_k = Hx_k + v_k \quad (2)$$

where:

- x_k = state vector,
- z_k = measurement vector,
- A, B, H = system matrices,
- u_k = control input,
- w_k, v_k = process and measurement noise.

FDI Attack Model

A False Data Injection (FDI) attack modifies the measurements as:

$$z'_k = z_k + a_k$$

where a_k is the attack vector. If properly designed, a_k can bypass traditional residual-based detection in WLS.

5. Kalman Filter Algorithm

The steps for implementation are as follows:

1. **Initialization:**

Set $\hat{x}_{0|0}$ (initial state) and $P_{0|0}$ (initial covariance).

2. **Prediction:**

$$\hat{x}_{k|k-1} = A\hat{x}_{k-1|k-1} + Bu_k$$

$$P_{k|k-1} = AP_{k-1|k-1}A^T + Q$$

3. **Measurement Update:**

$$r_k = z'_k - H\hat{x}_{k|k-1}$$

$$K_k = P_{k|k-1}H^T(H P_{k|k-1}H^T + R)^{-1}$$

4. **State Update:**

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k r_k$$

$$P_{k|k} = (I - K_k H)P_{k|k-1}$$

5. **Detection:**

If $|r_k| > \text{Threshold} \rightarrow \text{Raise Cyber Attack Alarm.}$

6. Algorithm Implementation Flow

1. Input IEEE 5-Bus system data.
2. Compute power injections and flows using Newton–Raphson method.
3. Add Gaussian noise to simulate measurement uncertainty.
4. Perform **WLS state estimation** (pre-attack, static case).
5. Set detection threshold using chi-square inverse CDF:

$$\chi_{\text{inv}}^2(1 - \alpha, m - n)$$

6. Apply stealth attack vector:

$$[0, 0, 0.577, -0.577, 0, 0, -0.577, 0, 0, 0]$$

7. Perform **Kalman Filter-based dynamic state estimation**.

8. Set threshold for dynamic model:

$$\chi_{inv}^2(1 - \alpha, m)$$

9. If residual crosses threshold $\rightarrow$ attack detected.

7. IEEE 5 Bus System

EE-513 Lab Assignment 7 (State Estimation)

Deadline: 19-10-2025

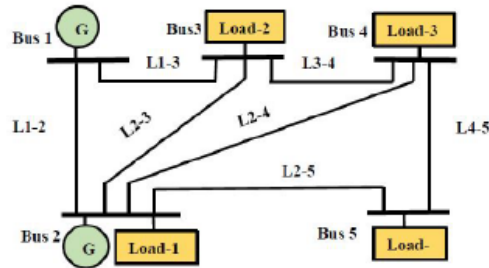


Figure 1: Schematics of IEEE 5-bus Network

Problem 1

Write a **MATLAB CODE** for AC State Estimation for IEEE 5-bus system shown in Fig. 1 by taking the required measurements from the model simulated in **PSCAD** and adding the White Gaussian Noise with following standard deviations (SD):

Measurement Type	Power Injections	Power Flows
Standard Deviation	0.004	0.008

Figure 1: IEEE 5-Bus System used for simulation.

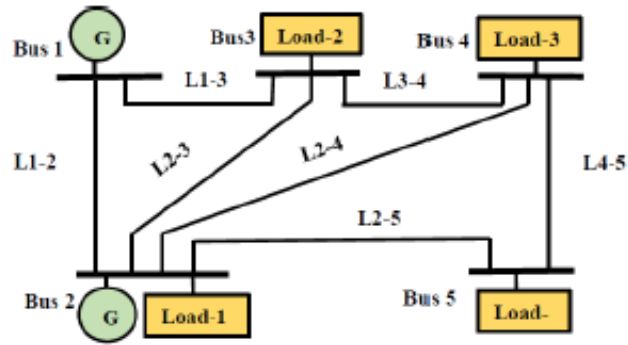


Fig.1. IEEE-5-Bus Network

Transmission line parameters of IEEE-5-Bus are given in per unit value in Table 1.

Table 1. Transmission line parameters of IEEE-5-Bus in per unit value.

From Bus	To Bus	Transmission line Resistance	Transmission line Reactance	Transmission line Conductance	Transmission line Susceptance
1	2	0.02	0.06	0	0.06
1	3	0.08	0.24	0	0.05
2	3	0.06	0.18	0	0.04
2	4	0.06	0.18	0	0.04
2	5	0.04	0.12	0	0.03
3	4	0.01	0.03	0	0.02
4	5	0.08	0.24	0	0.05

Table 3. Generation, load at the bus in per unit values

Bus	P_G	Q_G	Q-Max	Q-Min	P_L	Q_L
1	0	0	5	5	0	0
2	0.4	0	3	-3	0.2	0.1
3	0	0	0	0	0.45	0.15
4	0	0	0	0	0.4	0.05
5	0	0	0	0	0.6	0.1

If you want, you can ignore Q-max and Q-min limit for your convenience.

Table 5. Bus-type, initial voltage, and angle at the IEEE-5-bus network

Bus No.	1	2	3	4	5
Bus Type	Slack	PV	PQ	PQ	PQ
Initial V-magnitude	1.06	1	1	1	1
Initial V-angle	0	0	0	0	0

Figure 2: IEEE 5-Bus System used for simulation.

8. Results and Discussion

Key Observations

- Pre-attack thresholds:

$$\text{WLS Threshold} = 12.111, \quad \text{KF Threshold} = 31.1157$$

- As we see in fig.(3), The stealth attack successfully bypasses the WLS residual-based detection.
- As we see in fig.(4), The Kalman Filter detects the attack as the residual exceeds the dynamic threshold.
- As we see in fig.(5), The bus voltages deviated after the attack.

Residual Of WLS

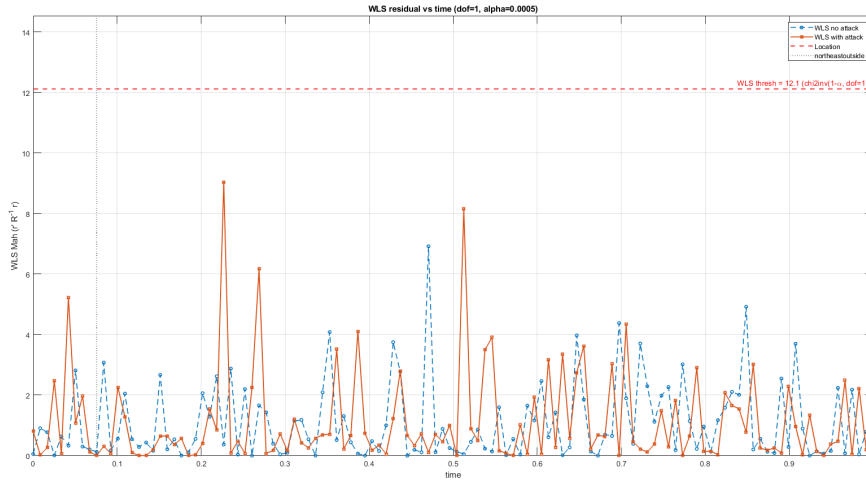


Figure 3: Residual of WLS vs Time

Residual of Kalman Filter

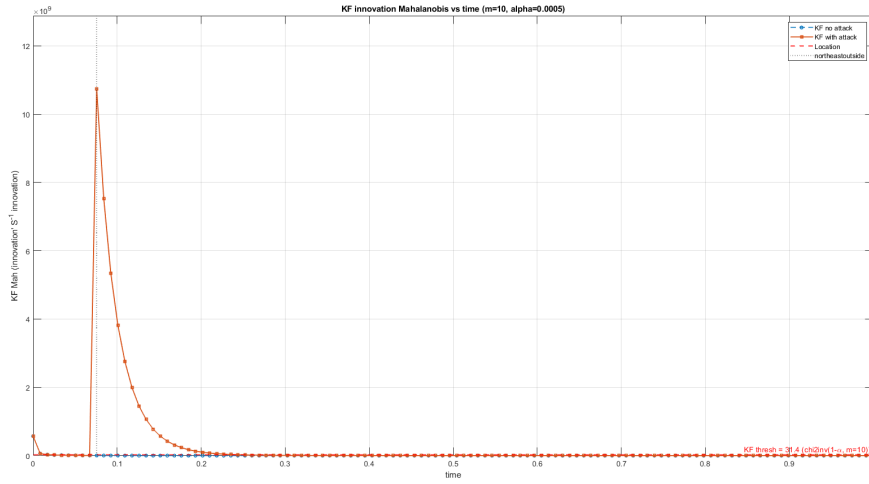


Figure 4: Residual of Kalman Filter vs Time showing attack detection.

Bus Voltages

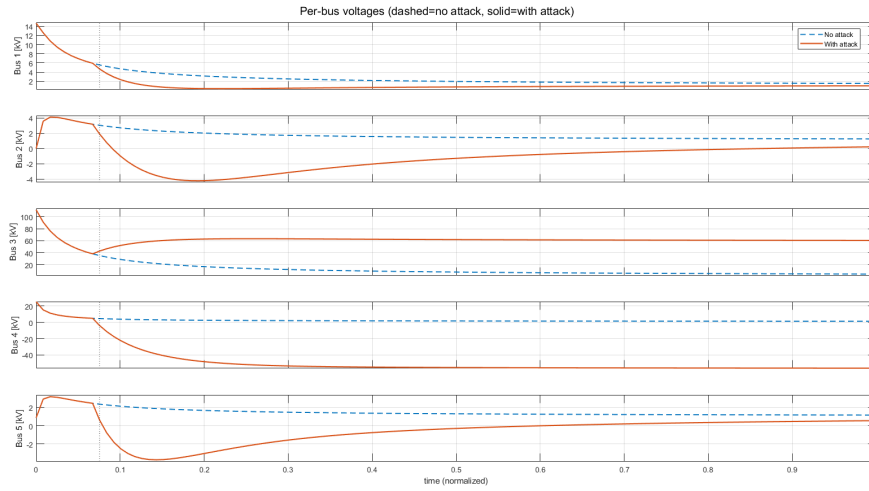


Figure 5: Bus Voltages vs time

9. Conclusion

- The Kalman Filter extends the WLS concept into a dynamic, time-recursive estimator.
- WLS provides only static estimation, while Kalman Filter continuously updates states.

- The Kalman Filter is more robust against stealth FDI attacks in smart grid systems.
- It efficiently detects anomalies by comparing measured and predicted data through residual analysis.

10. Reference

1. Yifa Liu, Long Cheng, “Relentless False Data Injection Attacks Against Kalman-Filter-Based Detection in Smart Grid,” *IEEE Transactions on Control of Network Systems*, Vol. 9, No. 3, pp. 2325–5870, Sept. 2022.